

Uday Bhaskar Kale

AI/ML Engineer - LLMs, RAG, Agentic Systems | M.Sc. Cognitive Systems (AI & ML), University of Potsdam

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Open to full-time AI/ML/LLM engineering roles

Education

2024 - 2026 (expected)	M.Sc. Cognitive Systems University of Potsdam, Germany Focus: NLP, Machine Learning, AI Agents, Multimodal Learning
2017 - 2021	B.E. Computer Engineering, University of Pune, India CGPA: 8.93 / 10

Professional Experience

Jan 2023 - May 2024	Technical Lead / Co-Founder, Seerlink Analytics Pvt Ltd, India - Built and deployed two production mobile applications (Flutter, Firebase) for dairy supply-chain management, enabling daily data collection, automated aggregation, and farmer-facing reporting - Designed a role-based data pipeline for milk collection records with validation, audit trails, and monthly summaries - Led end-to-end product delivery as co-founder, from requirements and system architecture to deployment and post-launch support
Aug 2021 - Dec 2022	Application Development Analyst, Accenture, India - Developed and maintained SAP PI/PO integrations across multiple enterprise systems in production environments - Owned data mapping, interface orchestration, and incident resolution for integration workflows - Supported testing, deployment, and production issue handling for business-critical systems - Collaborated with cross-functional teams to improve system reliability and data consistency

Technical Projects

KAKAPO - NL2SQL Corpus Query & Annotation Platform (Production System)	Python, FastAPI, SQLite, Docker, ChromaDB, spaCy, Groq API - Built and deployed a production NL2SQL platform that lets researchers query and annotate large text corpora in plain English, with an LLM translating natural language into SQL - Designed a privacy-first architecture where the LLM only ever receives the database schema, never the underlying data, removing any exposure of sensitive research data - Shipped with role-based access control, automated backup/restore, and Docker-based reverse-proxy/HTTPS deployment; live in production serving real users at the University of Potsdam
Argument-Aware Fact Checking	Python, SentenceTransformers, DeBERTa-v3, Qwen, Cerebras, LoRA - Built a fact-checking system that scores claims using argumentative evidence rather than keyword overlap, beating a flat-RAG baseline by +0.26 macro-F1 (0.388 vs. 0.127) on the LIAR-Arg benchmark - Distilled the reasoning into a 1.5B local model that retains 85% of a 120B-parameter teacher's accuracy, enabling fully offline inference with no dependency on cloud LLMs

	• Delivered as a reproducible artifact: fixed seeds, published model checkpoints on Hugging Face
Turix – Execution Authority Layer for AI Systems	Python, FastAPI, Apache-2.0 • Designed a policy-driven execution authority layer (control plane) for AI agent systems, enforcing ALLOW / DENY / ESCALATE decisions before real-world actions execute • Implemented tenant-aware authorization with immutable decision receipts, idempotent APIs, and fail-closed guarantees for safe agentic deployment • Architected strict separation between decision-making (AI) and execution authority (Turix), targeting domains such as payments, IAM, and infrastructure
Commit Message-Based Bug Localization	Python, scikit-learn, LangGraph, LLaMA (Groq) • Built multi-label classifier mapping commit messages → source files (~11k pandas commits, 20 targets) • Achieved Precision@1: 0.55 / Precision@3: 0.77 / Precision@5: 0.86 • Integrated LLM reranking via LangGraph pipeline with positional bias mitigation
Internal Documentation RAG System	Python, FAISS, SentenceTransformers, FLAN-T5, Streamlit • Built RAG pipeline (ingestion → embedding → retrieval → generation) over technical documentation • Used e5-base embeddings with FAISS for semantic retrieval • Enforced strict grounding with refusal when context unavailable • Provided answer + source attribution via interactive UI
Medical Triage AI Agent (Decision Support System)	Python, DistilBERT, LLM Policy (Groq) • Built an AI agent for clinical triage decision support (monitor / ask / escalate), explicitly avoiding diagnostic outputs • Designed pipeline combining ML classification, token-level evidence extraction, and LLM-based policy reasoning • Implemented human-in-the-loop compatible decision system with follow-up question generation • Structured agent loop (perception → reasoning → action) with modular and debuggable components

Technical Skills	
Programming	Python, Java, SQL, NoSQL
AI / Machine Learning	Machine Learning, Deep Learning, NLP, Multimodal Learning, Explainable AI, AI Agents
Frameworks	PyTorch, Hugging Face Transformers, LangGraph, Scikit-learn, Streamlit
LLM Systems	RAG, Prompt Engineering, LLM Evaluation
Data	Pandas, NumPy, Feature Engineering, Model Evaluation
Systems & Domains	API Design, Data Pipelines, Backend Systems
Tools	Git, Docker, FAISS, SAP PI/PO, SAP Basis

Languages
English - Professional, German - A2 (currently learning), Hindi - Native, Marathi - Native